[INCH-POUND]
A-A-59224
June 1, 1998
SUPERSEDING
GG-M-2802
July 15, 1993

COMMERCIAL ITEM DESCRIPTION

METERS, FLUID QUANTITY VOLUMETRIC

The General Services Administration has authorized the use of this commercial item description for all Federal agencies.

1. SCOPE. This commercial item description (CID) covers volumetric meters comprising of measuring elements that meter the fluid or gas in successive and individual quantities and registering elements that indicate the total fluid quantity passing through the meter since the last reading.

2. CLASSIFICATION. Meters covered by this CID are classified by type, indicating fluid or gas for which meter is designed, and style, indicating type of metering action, in accordance with the following (see 7.2):

TYPE

Type C	-	Condensate usage.
Type F	-	Fuel usage.
Type G	-	Gas usage.
Type W	-	Water usage.

STYLE

Style A	-	Oscillating piston or nutational disk.
Style B	-	Bellows or diaphragm.
Style C	-	Rotating drum.
Style D	-	Rotary impeller.

Beneficial comments, recommendations, additions, deletions, clarifications, etc., and any data which may improve this document should be sent to: Officer in Charge (Code 15E2), Seabee Logistics Center, 4111 San Pedro St., Port Hueneme, CA 93043-4410, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6680

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3. SALIENT CHARACTERISTICS.

3.1 Description. The meters shall consist of a volumetric-type measure drum, or a fixed displacement device, and a mechanically actuated registering element. The assembly that comprises the meter(s) shall be equipped for installation in a specified piping system. The fluid shall be measured by isolating the flow in a predetermined volume space or in a fixed displacement mechanism, such as a piston assembly, nutational disk assembly, bellows, diaphragm assembly, or rotary impeller. Meter parts in contact with the measured fluid shall be composed of material that will not chemically react with the fluid. When specified (see 7.2), meters (except gas meters) shall be equipped with a resettable counter, which shall be either part of the registering or integrating element, or provided as a separate unit. It shall indicate the accumulated fluid quantity that has passed through the meter since the last reading. Meter parts shall be readily accessible for maintenance and replacement, and shall be arranged to operate without vibration within the accuracy requirements contained herein. Meters, except those used for gas and condensate service, shall be used in feed water and distributing systems where water flow is 1,000 gallons per minute (gpm) (63.1 liter per second (L/s)) at temperatures up to 250 degrees Fahrenheit (°F) (121 degrees Celsius (°C)) at 150 pounds per square inch (psi) (1 034 kilopascal (kPa)). Meters shall be used in fuel feeding systems for oil flow of 660 gpm (41.6 L/s) at 150 psi (1 034 kPa) and normal operating temperature of 150 °F (65.5 °C). Gas meters are used in systems designed for maximum working pressures at 100 psi (689 kPa) and flow rates up to 10,000 cubic feet per hour (283 cubic metre per hour). Condensate meters are used to measure condensate flow at atmospheric pressure at rates up to 12,000 pounds per hour (1.51 kilogram per second (kg/s)) at 300 °F (149 °C). Actual range of operating conditions shall be as specified (see 7.2).

3.2 Design. The liquid meters shall be designed to operate without leakage or damage to any part at maximum pressure of 150 psi (1 034 kPa). The gas meters shall be designed to meet or exceed the requirements detailed in ANSI B109.1, B109.2 or B109.3 as appropriate for the meter type and capacity. The meters shall be designed for continuous operation, ease of installation and service in the field, constructed to prevent parts from working loose in service, and elimination of conditions hazardous to personnel and detrimental to the equipment. Instrument piping shall conform to ANSI B31.1. The meters shall be designed for fluid properties and system conditions specified (see 7.2). Adequate provisions shall be furnished to prevent corrosive liquid, air, foreign particles, or condensation from entering the indicating circuit of the instrument. When the operating temperature of steam or service liquid is greater than 240 °F (115 °C), condensation pots shall be provided where suitable.

3.3 Standard commercial product. The meters shall, as a minimum, be in accordance with the requirements of this CID and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this CID but which are a part of the manufacturer's standard commercial product, shall be included in the meters being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements or manufacturer's catalogs or brochures, and represents the latest production model.

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3.4 Interchangeability. All units of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to ensure interchangeability of component parts, assemblies, accessories, and spare parts.

3.5 Construction. The meters shall be constructed to maintain accuracy of indication and to resist wear and corrosion. Materials for construction of pressure-containing parts shall be selected in accordance with the system fluid and the system operating temperature and pressure. Moving parts shall be properly aligned, balanced, and lubricated. Interior parts, seals, and packings shall be accessible for service. Provisions for mounting either in the pipeline or on a pedestal shall be as specified (see 7.2). When specified (see 7.2), protection from meter overrange and reverse flow shall be furnished. A drain for fluid-containing components shall be provided. Construction shall be such that the measuring element or fixed displacement device shall cover the specified range without need for replacement.

3.6 Components. The meter shall consist of the following:

a. Accessory equipment as specified (see 7.2).
b. Additional apparatus normally furnished with these types of meters.
c. The components specified in 3.6.1 through 3.6.8.

3.6.1 Casing. The outer casing material shall be protected by suitable noncorrosive treatment, or steel that has been coated for purposes of corrosion-resistance. The case shall be dust-tight and designed for easy removal of interior parts. The gas meter casing shall be gas-tight and meet the requirements of ANSI B109.1, B109.2 or B109.3.

3.6.2 Measuring chamber, drum, or bellows. The measuring chamber, drum, or bellows shall maintain a constant volume displacement or constant volumetric measurement with each cycle of the meter. The measuring equipment shall be corrosion-resistant material and shall have a high resistance to temperature change or to deterioration resulting from moisture. The sensitivity or accuracy of the measuring component shall not be adversely affected by slight distortion of the outer casing. The bellows shall be pliable and gas-tight. The chamber or drum shall be balanced to prevent misalignment. When the fluid is specified for measurement under variable temperature conditions, means shall be provided to compensate for variations in fluid density or viscosity. Transmission of the metered action shall be accomplished directly by mechanical means, with minimum friction and lost motion.

3.6.3 Fixed displacement measuring device. Piston or disk-type metering elements shall be supported and aligned to prevent distortion. Moving parts shall be kept to a minimum. The device shall be designed to provide and maintain accuracy. Provisions for correction of variances in fluid temperatures and reversal of flow, when applicable, shall be furnished. The element material shall be inherently capable of resisting corrosion. Piston nutations or oscillations shall be kept to a metering action, with minimum friction and lost motions and shall be accomplished by direct mechanical means.

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3.6.4 Register. When specified (see 7.2), a direct-reading, adjustable (except gas meters) register or index shall be furnished. Materials shall be selected that resist corrosion. The register shall be graduated in the units specified, of the single or multiple dial type as applicable, and calibrated at center scale unless otherwise specified (see 7.2). The operating mechanism shall be designed for minimum lost motion, pressure drop, or friction and shall be encased in a dust-proof housing. The scale shall be clearly visible at a distance of 10 feet (3.05 metre (m)). Normal system variations or slight distortion of the housing shall not be subjected to undue strain as a result of the action of the transmitting mechanism.

3.6.5 Totalizer. A combined register-totalizer or a separate integrator shall be furnished, as specified (see 7.2). Materials shall be resistant to deterioration resulting from moisture or fog on the dial glass. The counter shall clearly indicate the cumulative metering actions. When specified (see 7.2), a water escape hole shall be provided in the housing. Slight distortion of the housing shall not cause the totalizer to fail. When specified (see 7.2), a record of the accumulated flow rate may be prepared by means of an electrically activated recorder. The total digit wheels to be provided shall be based on the time interval between readings, as specified (see 7.2), and the size meter selected.

3.6.6 Piping. Unless otherwise specified (see 7.2), the meters shall be furnished complete with integral piping for accessory equipment and for installation into required pipeline. When specified (see 7.2), a system strainer shall be supplied.

3.6.7 Bearings. Bearings provided in any of the meter components shall be of the antifriction type, lifetime-lubricated, and suitable for long life and resistance to wear. Bearing selection shall be based on the accuracy of fit required, ease of replacement, and the ability to maintain true alignment.

3.6.8 Air-eliminator. When specified (see 7.2), an air-eliminator shall be furnished.

3.6.9 Tamper-proofing. When specified (see 7.2), a tamper-proof seal shall be provided.

3.6.10 Freeze protection. When specified (see 7.2), meters (except gas meters) shall be furnished with a freeze-protective device, designed to yield to break under normal freezing conditions in order to reduce damage to other meter parts.

3.6.11 Identification markings. Identification shall be permanently and legibly marked directly on the meter or on a corrosion resisting metal plate securely attached to the meter at the source of manufacturer. Identification shall include the item name, the part or identifying number, the manufacturer’s model number and serial number, and the manufacturer’s name or trademark to be readily identifiable to the manufacturer.

3.6.12 Repair parts and service. Replacement or repair parts and service for the standard commercial product provided under this CID shall be available from the manufacturer, through commercial parts distribution, or service organizations.

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3.6.13 Instruction manual. The contractor shall furnish manufacturer's standard manuals normally furnished in commercial practice for each meter specified in the contract.

4. REGULATORY REQUIREMENTS.

4.1 Materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR). Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this commercial item description are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. Unless otherwise specified, none of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this commercial item description.

4.2 Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within specified tolerances using conversion tables contained in the latest version of ASTM SI-10 (IEEE/ASTM SI-10), and all other requirements of this commercial item description including form, fit, and function are met. If a product is manufactured to metric dimensions and these dimensions exceed the tolerances specified in the inch-pound units, a request should be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

5. QUALITY ASSURANCE PROVISIONS.

5.1 Product conformance. The products provided shall meet the salient characteristics of this CID, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial market. The government reserves the right to require proof of such conformance.

6. PACKAGING. The preservation, packing, and marking shall be as specified in the contract or order.

7. NOTES.

7.1 Source of documents.

7.1.1 The Federal Acquisition Regulation (FAR) and Code of Regulations (CFR) may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

7.1.2 ANSI Standards are available from the American National Standards Institute, 11 West 42nd Street, New York, NY 10036.

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7.1.3 ASTM Standards are available from the American Society of Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.

7.1.4 IEEE Standards are available from the Institute of Electrical and Electronics Engineers (IEEE), IEEE Service Center, 445 Hoes Lane, P.O. Box 1331, Piscataway, NJ 08855-1331.

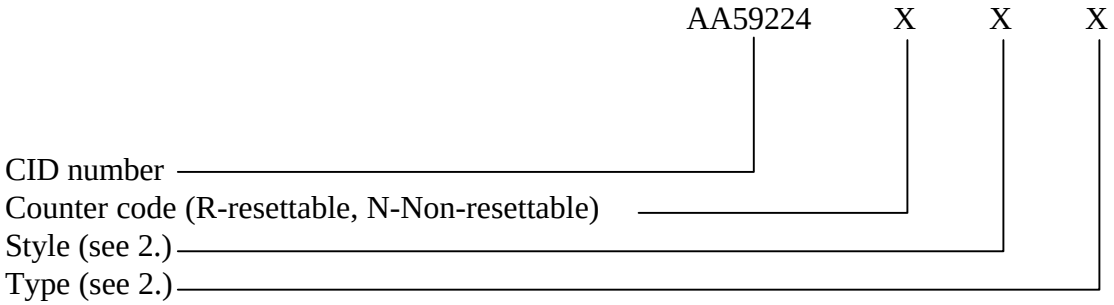
7.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this CID.
b. Type and style meter required (see 2. and 3.1).
c. When a resettable counter is required to indicate the accumulated fluid quantity since the last reading (see 3.1).
d. Density and viscosity of fluid to be metered, system operating temperature, and pressure flow range (3.1 and 3.2).
e. Whether pipe or pedestal mounting is required, and if meter overrange and reverse flow protection is required (see 3.5).
f. Accessory equipment required, such as gages and alarms (see 3.6).
g. When register is required, units required, dial type, and calibration if other than specified (see 3.6.4).
h. Type of totalizer required, when a water escape hole is required in the housing, when the accumulated flow rate may be prepared by an electrical recorder; time intervals between readings (see 3.6.5).
i. Piping connections required, and strainer if required (3.6.6).
j. When an air eliminator is required (see 3.6.8).
k. When meter is to be tamper-proof (see 3.6.9).
l. When a freeze-protective device is furnished (see 3.6.10).

7.3 Supersession data. This CID replaces Federal Specification GG-M-2802, dated July 15, 1993.

7.4 Classification cross reference. Classifications used in this CID (see 2.) are identical to those found in superseded Federal Specification GG-M-2802.

7.5 Part Identification Number (PIN). The following part identification numbering procedure is for government purposes and does not constitute a requirement for the contractor. The PIN to be used for items acquired to this description are created as follows:



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7.6 Metric units. The values stated in either inch-pound units or SI units are to be regarded separately as standard. Within the text, the SI units are shown in brackets. The values stated in each system are not exact equivalents; therefore, each system should be used independently of the other. Combining values from the two systems may result in non-conformance with the standard.

7.7 Subject term (key word) listing.

Condensate
Fuel
Gas
Water

MILITARY INTERESTS:

Custodian:
Navy - YD1
Army - AV

Review Activities:
Navy - AS
DSC -GS

CIVIL AGENCY COORDINATING ACTIVITY:

GSA-FSS

Preparing Activity:
Navy - YD1

(Project 6680-0256)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** A-A-59224	**2. DOCUMENT DATE (YYMMDD)** 980518

3. DOCUMENT TITLE

METERS, FLUID QUANTITY VOLUMETRIC

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON *(if applicable)*	**7.DATE SUBMITTED** (YYMMDD)

8. PREPARING ACTIVITY

a. NAME RUSSELL REYNOLDS	b. TELEPHONE *Include Area Code)* (1) Commercial 805-982-5946 (2) AUTOVON 551-5946
c. ADDRESS *(Include Zip Code)* OFFICER IN CHARGE, SEABEE LOGISTICS CENTER CODE 15E2, 4111 SAN PEDRO ST. PORT HUENEME, CA 93043-4410	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** DEFENSE QUALITY AND STANDARDIZATION OFFICE 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22401-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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